



USER-CENTERED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION

INTERIM DEMO





CONTENTS



WHY



WHO



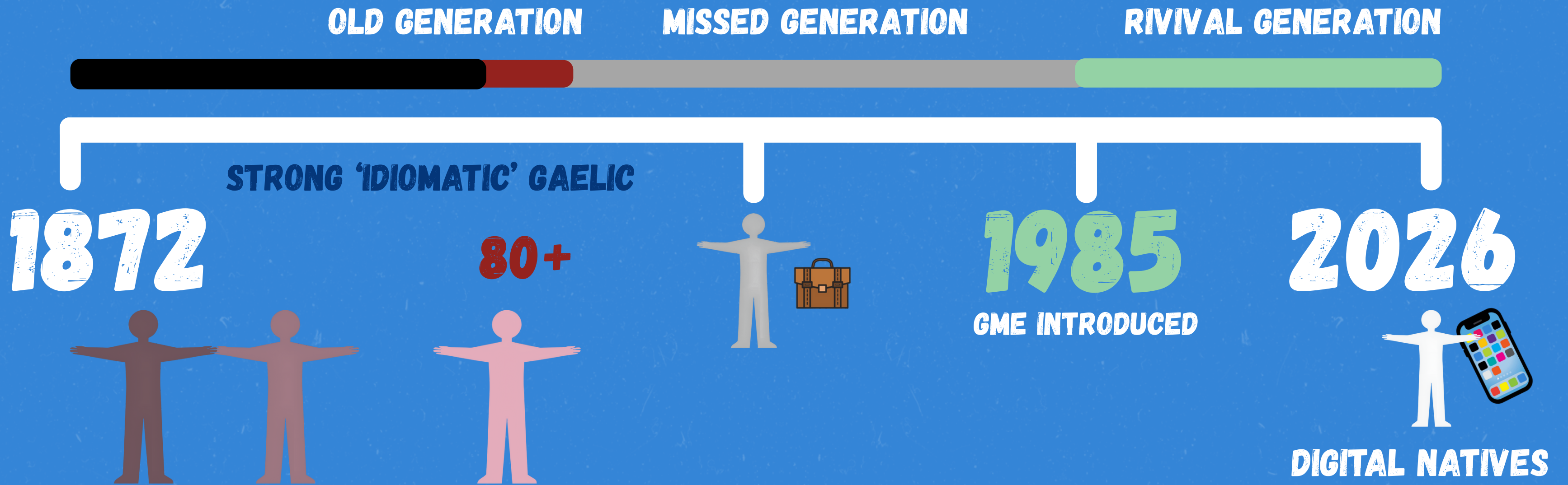
WHAT



HOW



WHY



Education Scotland Act 1872 excluded Gaelic from schools McLeod (2019)

By 2013-14, GME was available in 59 primary providers across 14 local authorities" O'Hanlon & Paterson (2015, p. 313)

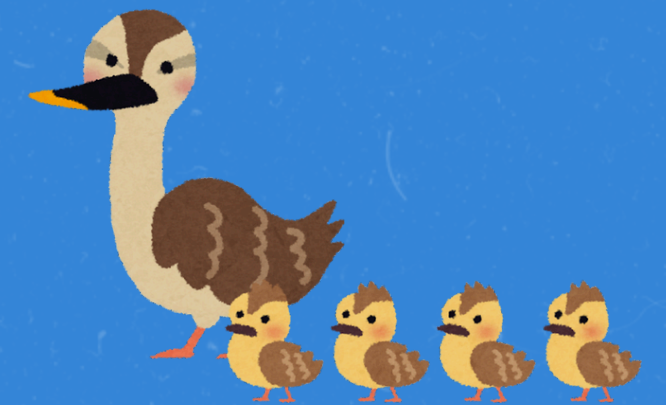
WHY

700,000 Scottish pupils getting digital devices by 2026 (Scottish Government) - without Gaelic-first tools, the language is omitted similarly to 1872 (2021)



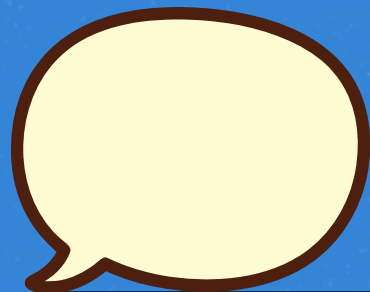
No such purpose-built Gaelic games Exist

Fishman (1991): Language revitalization requires "intergenerational home-family-neighbourhood transmission" - but many GME pupils come from non-Gaelic-speaking homes



Gaelic games deemed "commercially unviable" - Galloway (2011) concluded "cost would be prohibitive"

WHO



"Most school-educated children possess limited active and passive knowledge of **Gaelic idioms**" - NicLeid (2018)

P6-7 identified as "critical juncture where Gaelic proficiency begins to lag behind English. Pupils show "**reduced confidence in Gaelic-medium content**" at this transitional stage - O'Hanlon, McLeod & Paterson (2010)



Research Site: Bunsgoil Stèornabhaigh (Stornoway Primary School)

WHAT

A Multi-Game Gaelic Web Application

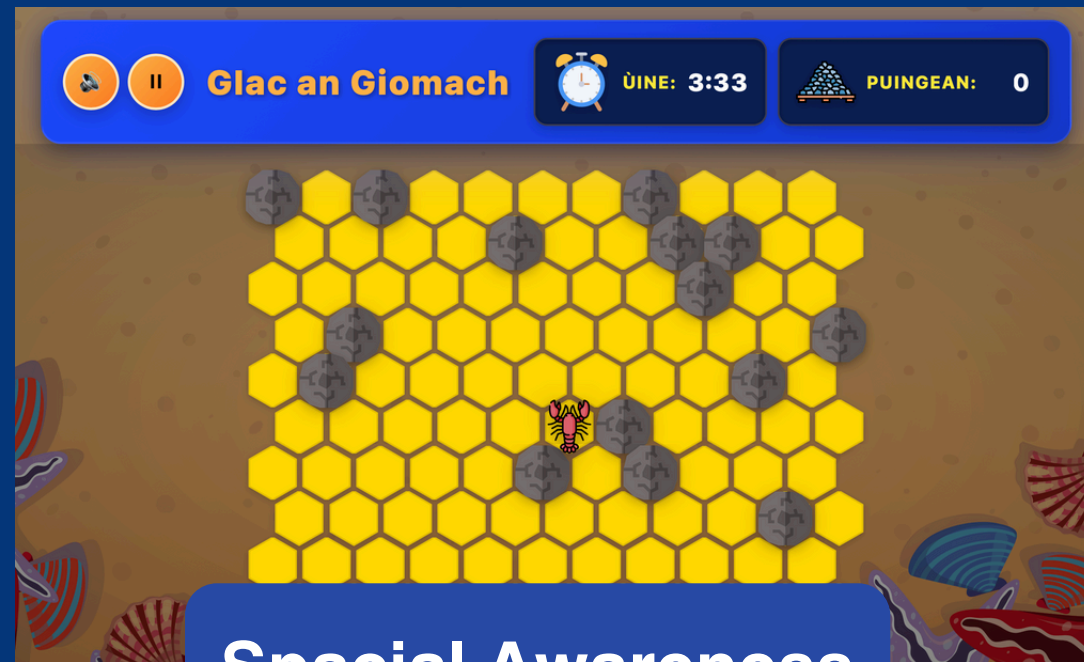
User Centred Design

Culturally authentic

100% Gaelic

Self-guided tutorials

GAME 1



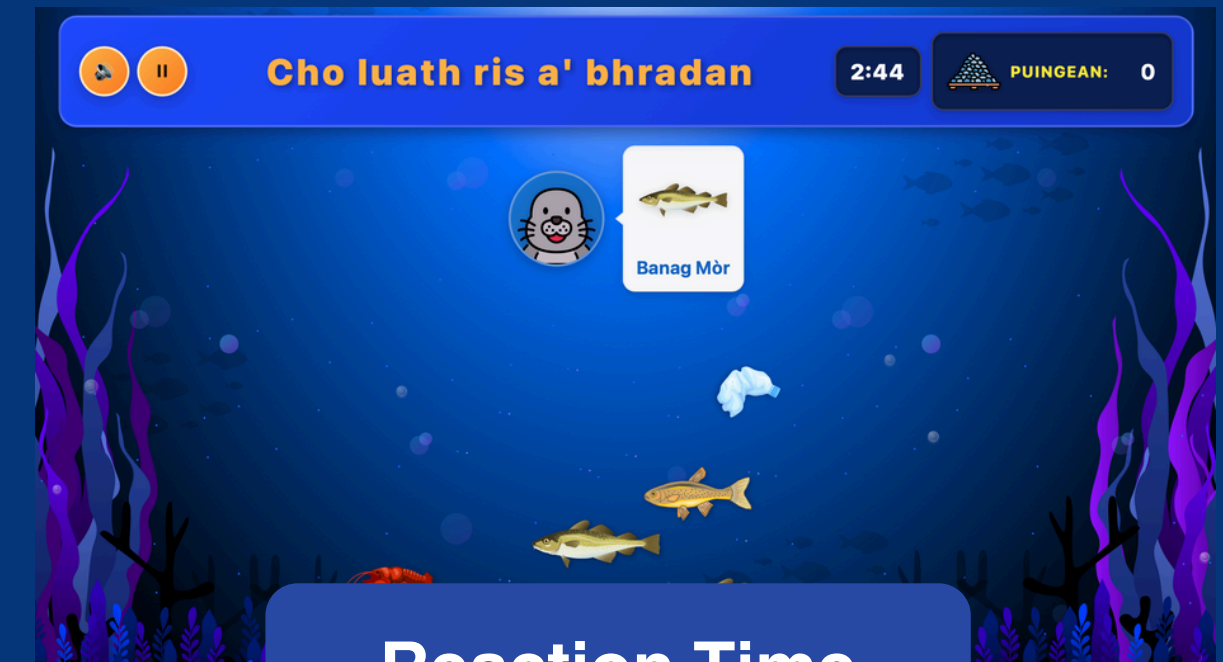
Spatial Awareness

GAME 2



Memory

GAME 3



Reaction Time

HOW



**Gaelic:
Past Present Future**



**Parallel Contexts:
Cross-Culture Design**



**Game Design:
Appropriate Frameworks**

Literature Review

HOW

User Centred Design (UCD):

Personas
User Profile
Stakeholder Meetings

Content Design:

Theme
Indexical Elements
Idiomatic Gaelic

Game Design Document (GDD):

Requirements
Games's Structure

Design

WATER



- LITERATURE REVIEW
- REQUIREMENTS
- METHODOLOGY
- ETHICS

WATER-SCRUM-FALL

(Butgereit, 2018, p. 177).

SCRUM



- CORE DEVELOPMENT
- SPRINTS
- BACKLOG FROM GDD
- ITERATIVE
- PROGRESS LOG

FALL



- EVALUATION
- IMPROVEMENTS
- REPORTING

Software Process

HOW

The game serves as both:
Educational tool - towards digital normalisation of Gaelic
Research instrument - vehicle for answering these 4 questions

Usability: To what extent is the game prototype usable for GME students aged 8-12 with minimal prior instruction?

Engagement & Experience: How do students and staff experience the game, and what are their perceptions of its cultural authenticity and engagement value?

Language Use: To what extent do pupils perceive the Gaelic in games as enjoyable and usable, and would they choose to engage with them? What linguistic factors support or hinder classroom adoption?

Emergent Phenomena: What unexpected barriers, facilitators, or phenomena emerge that were not anticipated by the design?

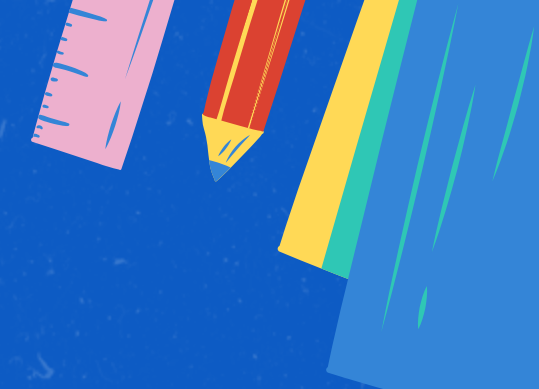
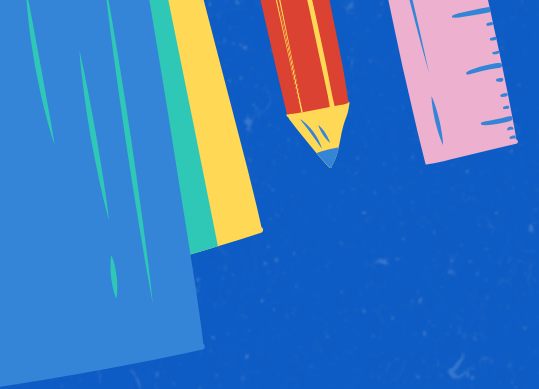
Research Output

Instrument	Question/Item	Data Type	Description	RQ1	RQ2	RQ3	RQ4
Pupil Questionnaire	SUS Item 1: Frequency of use	Quantitative	I think that I would like to use this system frequently	✓	✓		
Pupil Questionnaire	SUS Item 2: Complexity	Quantitative	I found the system unnecessarily complex	✓			
Pupil Questionnaire	SUS Item 3: Ease of use	Quantitative	I thought the system was easy to use	✓			
Pupil Questionnaire	SUS Item 4: Technical support needed	Quantitative	I think that I would need the support of a technical person to be able	✓			
Pupil Questionnaire	SUS Item 5: Integration	Quantitative	I found the various functions in this system were well integrated	✓			
Pupil Questionnaire	SUS Item 6: Inconsistency	Quantitative	I thought there was too much inconsistency in this system	✓			
Pupil Questionnaire	SUS Item 7: Learnability	Quantitative	I would imagine that most people would learn to use this system ve	✓			
Pupil Questionnaire	SUS Item 8: Cumbersome	Quantitative	I found the system very cumbersome to use	✓			
Pupil Questionnaire	SUS Item 9: Confidence	Quantitative	I felt very confident using the system	✓	✓		
Pupil Questionnaire	SUS Item 10: Learning required	Quantitative	I needed to learn a lot of things before I could get going with this sys	✓			
Pupil Questionnaire	Engagement items	Mixed	Self-reported engagement and satisfaction (adapted for age group)		✓		✓
Pupil Questionnaire	Language use items	Mixed	Items on language use and perceived authenticity of idiomatic expressions			✓	✓
Teacher Questionnaire	Engagement evaluation	Qualitative	Professional evaluations of pupil engagement		✓		
Teacher Questionnaire	Cultural fit evaluation	Qualitative	Professional evaluations of cultural fit		✓		
Teacher Questionnaire	Cultural authenticity assessment	Qualitative	Assessment of cultural authenticity of language and references		✓	✓	
Teacher Questionnaire	Observed engagement	Qualitative	Observations of pupil engagement during gameplay		✓		
Teacher Questionnaire	Observed language use	Qualitative	Observations of pupil Gaelic language use			✓	
Teacher Questionnaire	Open-ended responses	Qualitative	Open-ended responses capturing unexpected phenomena				✓
Pupil Discussion Group	Cultural interests	Qualitative	Discussion prompts on cultural interests		✓		
Pupil Discussion Group	Game habits	Qualitative	Discussion prompts on game habits and preferences		✓		
Pupil Discussion Group	Perceptions of cultural elements	Qualitative	Discussion of perceptions of cultural elements in the game		✓		
Pupil Discussion Group	Game preferences	Qualitative	Discussion prompts on game preferences		✓	✓	
Pupil Discussion Group	Language preferences	Qualitative	Discussion prompts on language preferences			✓	
Pupil Discussion Group	Enjoyment	Qualitative	Discussion of enjoyment and engagement with the game		✓		
Pupil Discussion Group	Gaelic use during play	Qualitative	Discussion of Gaelic language use during gameplay			✓	
Pupil Discussion Group	Unexpected aspects	Qualitative	Reflections on unexpected experiences or phenomena				✓
Field Notes				✓	✓	✓	✓



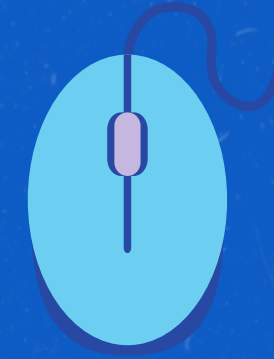
OBJECTIVES





OBJECTIVES

- **LITERATURE REVIEW**
 - **REQUIREMENTS FOR A GAELIC DIGITAL GAME**
 - **ELEMENTS/INTERACTION STRATEGIES FOR ACTIVE GAELIC USAGE**
 - **FUNCTIONAL PROTOTYPE**
- **IMPROVE ON PROTOTYPE THROUGH FEEDBACK**
 - **ANALYSE GAME BASED HCI PRINCIPLES TOWARDS CURRICULUM**
 - **IN GAME SUPPORT**
- **DOCUMENT REQUIRED ADAPTATIONS FOR FUTURE APPLICATIONS**
 - **INVESTIGATE APPLICABILITY TO OTHER MINORITY LANGUAGES**
 - **EXPLORE TEACHER-FACING CONFIGURATIONS**



THANK YOU

QUESTIONS WELCOME:

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